Selected Solutions from Tutorials Part 2

1. Adapted from **Shigley, Problem 12-22**.

   An Oiles SP 500 alloy brass bushing is 1 in long with a 1-in bore and operates in a clean environment at 70°F. The allowable wear without loss of function is 0.005 in. The radial load is 500 lbf. The shaft speed is 200 rev/min.

   Part 1: To be demonstrated by the TA.

   a. Estimate the number of revolutions for radial wear to be 0.005 in.

   Part 2: To be done by students.

   b. Estimate the wear expected for 1 million revolutions if the speed is reduced to 80 rev/min, the load is increased to 800 lbf, and the material is changed to Oiles SP 800.

Part 1: To be demonstrated by the TA.

a. Estimate the number of revolutions for radial wear to be 0.005 in.

The material removal rate (thickness per time) is:

$w = f_1 f_2 KFNt/(3L)$ (eq. 12-32, p. 664)

Note: eq. 12-27, p. 663 looks similar but is for flat surface wear, not curved-surface wear on a bushing.

where

P is the nominal pressure

$P = F/DL$ (eq. 12-28, p. 663)
$= (500\ lbf)/(1in \cdot 1in)$
$= 500\ psi$

V is the sliding velocity

$V = \pi DN/12$ (eq. 12-29, p. 664)
$= \pi(1in)(200\ rev/min)/12$
$= 52.36\ ft/min$

f_1 is the motion related factor

f_1 is given by Table 12-10 (p. 663)

Since the shaft is turning at 200 rev/min (constant) we are dealing with rotary motion in the table. With P = 500 psi and V = 52.36 ft/min (both found above) we get f_1 = 1.3-1.8. Of this range, the more conservative estimate is 1.8. (Alternatively, we could interpolate between the range and we would find f_1 = 1.44.) For now, use the conservative estimate:

$f_1 = 1.8$

f_2 is the environmental factor

f_2 is given by Table 12-11 (p. 663)

$f_2 = 1.0$ (<140°F in clean environment → no foreign matter)

* All problems, equations and page numbers are based on Shigley, 9th edition.

K is the wear factor

K is given in Table 12-8 (p. 662)

For Oiles 500, $K = 0.6\times10^{-10}\ in^3\cdot min/(lbf\cdot ft\cdot hr)$

$K = 0.6\times10^{-10}$

$w = f_1 f_2 KFNt/(3L)$

$t = 3Lw / (f_1 f_2 KFN)$
$= 3(1in)(0.005\ in) / [(1.8)(1.0)(0.6\times10^{-10})(500\ lbf)(200\ rev/min)]$
$= 1389\ hrs$

At 200 rev/min operating for 1389 hrs, the total number of revolutions is given by:

$Revs = (200\ rev/min)(1389\ hrs)(60\ min/hr) = 1.67\times10^7\ rev$

Revs = 16.7 million revolutions

Part 2: To be done by students.

b. Estimate the wear expected for 1 million revolutions if the speed is reduced to 80 rev/min, the load is increased to 800 lbf, and the material is changed to Oiles SP 800.

The material removal rate (thickness per time) is:

$w = f_1 f_2 KFNt/(3L)$ (eq. 12-32, p. 664)

P is the nominal pressure ← increased

$P = F/DL$ (eq. 12-28, p. 663)
$= (800\ \text{lbf})/(1\text{in} \cdot 1\text{in})$
$= 800\ \text{psi}$

V is the sliding velocity ← decreased

$V = \pi DN/12$ (eq. 12-29, p. 664)
$= \pi(1\text{in})(80\ \text{rev/min})/12$
$= 20.94\ \text{ft/min}$

f_1 is the motion related factor

f_1 is given by Table 12-10 (p. 663)

From the table with P = 800 psi and V = 26.18 ft/min, the values of f_1 range from 1.5 to 2.0. Take 2.0 as a conservative approach in wear.

$f_1 = 2.0$

f_2 is the environmental factor

f_2 is given by Table 12-11 (p. 663)

$f_2 = 1.0$ (<140°F in clean environment → no foreign matter)

K is the wear factor

K is given in Table 12-8 (p. 662)

For Oiles 800, $K = 3\times10^{-10}\ \text{in}^3\cdot\text{min}/(\text{lbf}\cdot\text{ft}\cdot\text{hr})$

$K = 3\times10^{-10}$

Number of revolutions = speed · time = n·t = 1 million/60 (given)

$w = f_1 f_2 KFNt/(3L)$
$= (2.0)(1.0)(3\times10^{-10})(800\ \text{lbf})(10^6/60)/(3)(1)$
$= 2.67\times10^{-3}\ \text{in}$

We should expect wear of 2.67×10^{-3} inches

2. Adapted from **Shigley, Problem 11-6** (8th Ed.).

An 02-series deep-groove ball bearing is to be selected to carry a radial load of 8kN and a thrust load of 4 kN. The desired life, L_D, is to be 5000 hours with an inner-ring rotation rate of 900 rev/min.

Part 2: To be done by students.

b. What is the basic load rating that should be used for an angular contact bearing if the outer-ring rotates at 1000 rev/min, the axial load is increased to 5 kN, and a reliability goal of 0.95 is selected?

From the question

F_r = 8kN (radial load)
F_a = 5 kN (axial load)
L_D = (5000 hrs)(1000 rev/min)(60 min/hr) (design life)
= 300 million rev
V = 1.2 (rotation factor, see 1st paragraph, Sec. 11-6, p. 579)

Need to combine radial and trust loads to an effective radial load.

$F_e = X_i V F_r + Y_i F_a$ From eq. 11-9 (p. 580)

The values of X_i and Y_i are determined from Table 11-1. We need to know what the static load rating, C_0, is to find a parameter e. The values of X_i and Y_i used depend on whether the quantity $F_a/(VF_r)$ is greater than or less than e. In order to find e, we need to use trial and error to select a bearing and use it's value of C_0.

First Trial

Considering the radial load only,

$$C_{10} = F_D \left[\frac{x_D}{x_0 + (\theta - x_0)\left(\ln \frac{1}{R_D}\right)^{\frac{1}{b}}} \right]^{\frac{1}{a}}$$ (eq. 11-6, p. 578)

F_D =8 kN (given in question)
x_D = L_D/L_{10}
= 300
x_0 = 0.02 (given in the question supplementary info)
θ = 4.459 (given in the question supplementary info)
b = 1.483 (given in the question supplementary info)
a = 3 (ball bearing: see below eq. 11-1, p. 574)
R_D = 0.95 (given in the question)

$$C_{10} = 8\ \text{kN} \left[\frac{300}{0.02 + (4.459 - 0.02)\left(\ln \frac{1}{0.95}\right)^{\frac{1}{1.483}}} \right]^{\frac{1}{3}}$$
$$= 62.8\ \text{kN}$$

From Table 11-2, the 65 mm bore angular contact ball bearing has C_{10} = 63.7 kN. We could start with this bearing, but as we saw above, the axial load is

going to increase the effective radial load and we will quickly find this bearing is too small. Instead, start with the next biggest bearing (from Table 11-2, 70 mm bore, C_{10} = 68.9 kN and C_0 = 45.5 kN).

Referring to Table 11-1 with F_a = 5 kN (given in question) and C_0 = 45.5 kN (found from Table 11-2)

$$F_a/C_0 = (5\text{ kN})/(45.5\text{ kN}) = 0.110$$

In Table 11-1, F_a/C_0 = 0.110 corresponds to a value of e of 0.30.

As instructed in the table, we compare $F_a/(VF_r)$ to e

$$F_a/(VF_r) = (5\text{ kN}) / (1.2 \cdot 8\text{ kN}) = 0.52$$

This is greater than e, so we need to use the values of $X_i = X_2 = 0.56$ and $Y_i = Y_2 = 1.45$.

From the equivalent load expression above,

$F_e = X_iVF_r + Y_iF_a$ From eq. 11-9 (p. 580)
$= (0.56)(1.2)(8\text{ kN}) + (1.45)(5\text{ kN})$
$= 12.63$ kN

Computing the new value of C_{10} based on this new equivalent radial load gives

$$C_{10} = 12.63\text{ kN}\left[\frac{300}{0.02 + (4.459 - 0.02)\left(\ln\frac{1}{0.95}\right)^{\frac{1}{1.483}}}\right]^{\frac{1}{3}}$$

$$= 99.2\text{ kN}$$

Comparing to our original bearing selection, the C_{10} value is 68.9 kN. Once the axial load is included, our bearing is no longer large enough. Looking back to the table, we see that a 90 mm bearing will work based on our current approximation of the rated load, C_{10} (C_{10} for a 90 mm bore bearing is 106 kN).

Second Trial

The new value of C_0 (from Table 11-2) is 73.5 kN. Performing the same calculations above, we

$$F_a/C_0 = (5\text{ kN})/(73.5\text{ kN}) = 0.068$$

In Table 11-1, F_a/C_0 = 0.056 corresponds to a value of e of 0.26; F_a/C_0 = 0.070 corresponds to a value of e of 0.27. Linear interpolation gives a value of e of 0.269.

$F_a/(VF_r)$ = 0.52 (unchanged from before)

This is greater than e, so we need to use the values of $X_i = X_2 = 0.56$ and $Y_i = Y_2 = 1.64$. (this value was found using linear interpolation)

From the equivalent load expression above,

$F_e = X_iVF_r + Y_iF_a$ From eq. 11-9 (p. 580)
$= (0.56)(1.2)(8\text{ kN}) + (1.64)(5\text{ kN})$
$= 13.58\text{ kN}$

Computing the new value of C_{10} based on this new equivalent radial load gives

$$C_{10} = 13.58\text{ kN}\left[\frac{300}{0.02 + (4.459 - 0.02)\left(\ln\frac{1}{0.95}\right)^{\frac{1}{1.483}}}\right]^{\frac{1}{3}}$$
$$= 106.6\text{ kN}$$

Alas, this bearing is just a hair too small (we have C_{10} = 106 kN for the bearing but need 106.6 kN based on the above calculation). We will need to use the 95 mm bore bearing. The confirmation is shown below.

Third Trial

The new value of C_0 (for the 95 mm bearing from Table 11-2) is 85.0 kN. Performing the same calculations above, we

$F_a/C_0 = (5\text{ kN})/(85.0\text{ kN})$
$= 0.0588$

In Table 11-1, $F_a/C_0 = 0.056$ corresponds to a value of e of 0.26; $F_a/C_0 = 0.70$ corresponds to a value of e of 0.27. Linear interpolation gives a value of e of 0.262.

$F_a/(VF_r)$ = 0.52 (unchanged from before)

This is greater than e, so we need to use the values of $X_i = X_2 = 0.56$ and $Y_i = Y_2 = 1.694$. (this value was found using linear interpolation)

From the equivalent load expression above,

$F_e = X_iVF_r + Y_iF_a$ From eq. 11-9 (p. 580)
$= (0.56)(1.2)(8\text{ kN}) + (1.694)(5\text{ kN})$
$= 13.87\text{ kN}$

Computing the new value of C_{10} based on this new equivalent radial load gives

$$C_{10} = 13.87\ \text{kN}\left[\frac{300}{0.02 + (4.459 - 0.02)\left(\ln\frac{1}{0.95}\right)^{\frac{1}{1.483}}}\right]^{\frac{1}{3}}$$

$$= 108.9\ \text{kN}$$

Finally, this value of C_{10} computed with the equivalent radial load remains within the limits of the actual bearing (108.9 kN < 121 kN).

Select a 02-series angular contact bearing with 95 mm bore bearing.

Adapted from **Shigley, Example 7-6** (8th and 9th Eds.) A steel shaft has a diameter of 2 in. The shaft rotates at 600 rev/min and transmits 5 hp through a gear. (to be done by students)

a. What size cup-point setscrew would be required to hold the gear. Note: a safety factor of 4 is a typical minimum for a setscrew subjected to dynamic loading.

b. What size dowel pin would be required for the same gear, if the pin passes through both sides of the shaft? (select a pin from the class notes) Assume a safety factor of 3 is required.

c. What size taper pin would be required for the same gear, assuming the pin is loaded in single shear? (select a pin from the class notes) Assume a safety factor of 3 is required.

d. Select an appropriate size gib-head key to hold the gear if the power transmitted is increased to 40 hp (assume the key is made of steel with yield strength of 70 kpsi). A safety factor of 2.5 is desired. Use the distortion-energy theory to select a key (note that under distortion-energy, the shear yield strength is $S_{sy} = 0.577S_y$).

1. Adapted from Shigley, Example 7-6 (8th and 9th Eds.)

A steel shaft has a diameter of 2 in. The shaft rotates at 600 rev/min and transmits 5 hp through a gear.

Part 1 - To be done by the students

a. **What size cup-point setscrew would be required to hold the gear. Note: a safety factor of 4 is a typical minimum for a setscrew subject to dynamic loading.**

The torque transmitted by the shaft, T, is related to power transmitted, H, and rotation rate, ω

$$T = H/\omega \qquad \text{(in SI units)}$$

When converted to torque in lbf·in, power in hp, and rotation rate in rev/min (where ω is replaced by n to acknowledge the change in units)

$$T = 63{,}025\ H/n$$
$$= 63{,}025\ (5\ \text{hp}) / (600\ \text{rev/min})$$
$$= 525\ \text{lbf·in}$$

For a 2" diameter shaft, the force at the surface is:

$$F = T / (d/2)$$
$$= 2T / d$$
$$= 2\ (525\ \text{lbf·in})/(2\ \text{in})$$
$$= 525\ \text{lbf}$$

Applying a safety factor of 4, the setscrew must support a load of 4 times the surface force = 2100 lbf. From Table 7-4, a 7/16" setscrew would be sufficient (a 7/16" setscrew provides a holding power of 2500 lbf). As a rule of thumb, set screw diameter should normally be about ¼ of the mating shaft diameter - this is the case here.

Select a 7/16" setscrew

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b. **What size dowel pin would be required for the same gear, if the pin passes through both sides of the shaft? (select a pin from the class notes) Assume a safety factor of 3 is required.**

From part a, the torque to be transmitted is

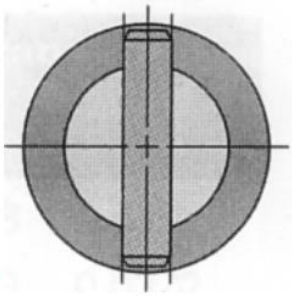
$$T = 525\ \text{lbf·in}$$

and the force at the shaft surface (for a 2" diameter shaft) is:

$$F = 525\ \text{lbf}$$

Applying a safety factor of 3, the pin must support a load of 3 times the surface force = 1575 lbf. The pin passes through the shaft, so it is loaded in double shear (see image below). From the table, a 0.125" diameter dowel pin in double shear can handle 3200 lbf.

Double shear:



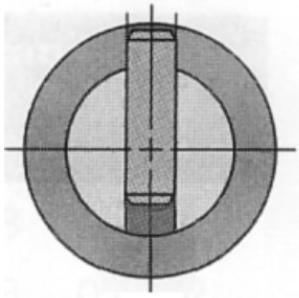
Select 0.125" = 1/8" diameter dowel pin

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c. What size taper pin would be required for the same gear, assuming the pin is loaded in single shear? (select a pin from the class notes) Assume a safety factor of 3 is required.

This is the same as part b, except now the taper pin is loaded in single shear. The numbers from the table must be reduced by two (or the required load doubled).

Single shear:



The force at the shaft surface (for a 2" diameter shaft) is:

F = 525 lbf

Applying a safety factor of 3, the pin must support a load of 3 times the surface force = 1575 lbf. From the table, a 0.125" diameter taper pin in double shear can handle 2900 lbf. In single shear, the pin an handle approximately 1450 lbf (it will actually be a bit more, since the wide end of the pin will be taking the load). Therefore, select a 0.188" diameter taper pin (6600 lbf in double shear, approx. 3300 lbf in single shear).

Select a 0.188" = 3/16" diameter taper pin

d. Select an appropriate size gib-head key to hold the gear if the power transmitted is increased to 40 hp (assume the key is made of steel with yield strength of 70 kpsi). A safety factor of 2.5 is desired. Use the distortion-energy theory to select a key (note that under distortion-energy, the shear yield strength is $S_{sy} = 0.577S_y$).

The new torque transmitted to the shaft is

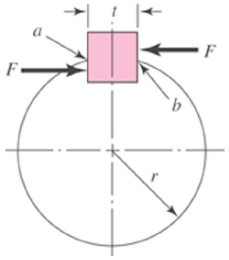
$$\begin{aligned} T &= 63{,}025\ H/n \\ &= 63{,}025\ (40\ hp)\ /\ (600\ rev/min) \\ &= 4200\ lbf\cdot in \end{aligned}$$

For a 2" diameter shaft, the force at the surface is:

$$\begin{aligned} F &= T\ /\ (d/2) \\ &= 2T\ /\ d \\ &= 2\ (4200\ lbf\cdot in)/(2\ in) \\ &= 4200\ lbf \end{aligned}$$

(Unlike the previous part, the convention for this case is to apply the safety factor to the shear stress prior to comparison with material strength. The answer would still work out correctly if the safety factor were applied to the above force, as was done in the previous part.)

This force places the key in direct shear as well as in compression.



The shear stress is $\tau = F/A_s$ where A_s is the area of the key between points a and b in the figure above. Specifically, $A_s = w\cdot l$ where w is the thickness of the key (shown as t in the figure) and l is the depth into the page. Both of these quantities need to be determined by selecting a key from Table 7-6.

$$\tau = F/(w\cdot l)$$

This stress must remain below the shear yield strength, adjusted by the safety factor.

$$\tau \le S_{sy}\ /\ n$$

$$F/(w\cdot l) \le S_{sy}\ /\ n$$

$$(w\cdot l) \ge F\cdot n/S_{sy}$$

The shear yield strength (as given in the question) is:

$$\begin{aligned} S_{sy} &= 0.577\ S_y \\ &= 0.577\ (70\ kpsi) \\ &= 40.4\ kpsi \end{aligned}$$

$$(l\cdot w) \ge (4200\ lbf)(2.5)\ /\ (40.4\ kpsi)$$

$$(l\cdot w) \ge 0.2600$$

In addition, the key must resist crushing by compression. The area, A_c, to be used in this case is the product of the depth of the key into the shaft with the shaft length, l. Standard practice (and as shown by comparing the last two columns of Table 7-6) is to have the key depth equal to one-half of the key height, h/2.

$$\sigma = F / A_c$$

$$F/(l \cdot h/2) \leq S_y / n$$

$$l \cdot h \geq 2F \cdot n / S_y$$

The compression yield strength is:

S_y = 70.0 kpsi

$$(l \cdot h) \geq 2(4200 \text{ lbf})(2.5)/(70 \text{ kpsi})$$

$$(l \cdot h) \geq 0.300$$

From Table 7-6, the recommended key size for a 2 in shaft (i.e. between 1-3/4" and 2-1/4") is either:

w×h = ½" × 3/8" or ½" × ½"

Either key size would work, as shown below.

For w×h = ½" × 3/8"

$l \cdot w \geq 0.26$
$l \geq 0.26 / 0.5$ in
≥ 0.52 in

$l \cdot h \geq 0.30$
$l \geq 0.30 / 0.375$ in
≥ 0.80 in

Minimum length, 0.80 in

For w×h = ½" × ½"

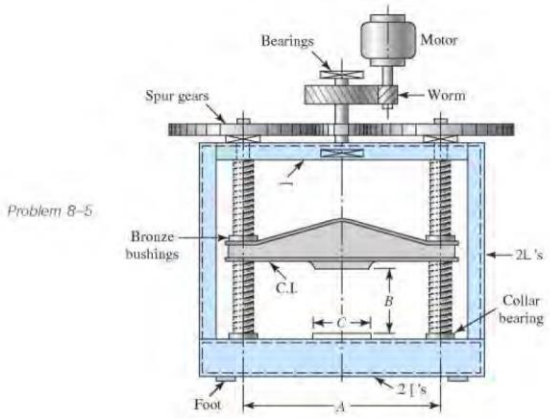
$l \cdot w \geq 0.26$
$l \geq 0.26 / 0.5$ in
≥ 0.52 in

$l \cdot h \geq 0.30$
$l \geq 0.30 / 0.5$ in
≥ 0.60 in

Minimum length, 0.6 in

Choose either a 0.80 in long ½"×3/8" key or a 0.60 in long ½"×½" key.

Problem 2. Adapted from **Shigley, Problem 8-6**.



The press shown above has a rated load of 5000 lbf. The twin screws have Acme threads, a diameter of 2 in, and a pitch of $\frac{1}{4}$ in. Coefficients of friction are 0.05 for the threads and 0.08 for the collar bearings. Collar diameters are 3.5 in. The gears have an efficiency of 95 percent and a speed ratio of 60:1. A slip clutch, on the motor shaft, prevents overloading. The full-load motor speed is 1720 rev/min.

Part 1: To be done by students.

(*a*) When the motor is turned on, how fast will the press head move?
(*b*) What should be the horsepower rating of the motor?

Screws rotate at an angular rate of

$$n = \frac{1720}{60} = 28.67 \text{ rev/min}$$

(a) The lead is 0.25 in, so the linear speed of the press head is

$V = 28.67(0.25) = 7.17$ in/min *Ans.*

(b) $F = 2500$ lbf/screw

$$d_m = 2 - 0.25 / 2 = 1.875 \text{ in}$$
$$\sec \alpha = 1 / \cos(29^\circ / 2) = 1.033$$

Eq. (8-5):

$$T_R = \frac{2500(1.875)}{2}\left(\frac{0.25 + \pi(0.05)(1.875)(1.033)}{\pi(1.875) - 0.05(0.25)(1.033)}\right) = 221.0 \text{ lbf} \cdot \text{in}$$

Eq. (8-6):

$$T_c = 2500(0.08)(3.5 / 2) = 350 \text{ lbf} \cdot \text{in}$$
$$T_{total} = 350 + 221.0 = 571 \text{ lbf} \cdot \text{in/screw}$$
$$T_{motor} = \frac{571(2)}{60(0.95)} = 20.04 \text{ lbf} \cdot \text{in}$$
$$H = \frac{Tn}{63\,025} = \frac{20.04(1720)}{63\,025} = 0.547 \text{ hp} \quad \textit{Ans.}$$

Problem 2 (Mott 12-23)

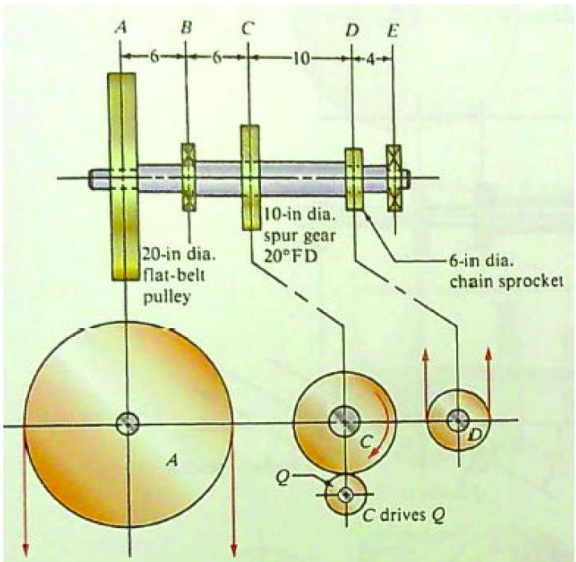


FIGURE P12–2 (Problems 2, 12, 13, and 23)

The shaft shown in the Figure is rotating at 200 rpm. The shaft carries a spur gear at C having 80 teeth with diametrical pitch of 8. The teeth are 20° full depth, involute form. The shaft carries a 20 inch diameter flat belt pulley and receives 10 HP from below at A. The shaft delivers 6 HP to the mating gear at C. The chain sprocket at D delivers 4HP the shaft above. The shaft is supported by bearings at B and E.

A) Determine the forces acting on the shaft and the reactions at the bearings.
B) Draw the complete load, shear, bending moment, torque diagrams for the shaft.
C) Specify the diameter of the shaft at the spur gear location. Assume the shaft is machined from SAE 1117 cold-drawn steel. Assume a key way is used to deliver the torque to the gear and a *ring groove* is machined in the shaft at the right of the gear to support a retaining ring.

Solution

23. Figure P12-2

Torque on Pulley *A*: $T_A = 63\,000(10)/200 = 3150$ lb · in

Torque on Gear *C*: $T_C = 63\,000(6)/200 = 1890$ lb · in

Torque on Sprocket D: $T_D = 63\,000(4)/200 = 1260$ lb · in

Torque Distribution in Shaft: $T_{A-C} = 3150$ lb · in; $T_{C-D} = 1260$ lb · in; $T_{D-E} = 0$

Forces on Pulley *A*:

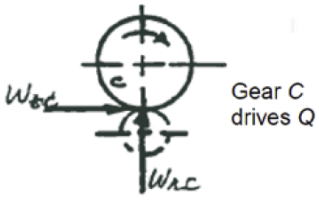
$$F_1 - F_2 = \frac{T_A}{R_A} = \frac{3150}{10} = 315 \text{ lb}$$

$$F_1 + F_2 = 2.0(F_1 - F_2) = 2.0(315) = 630 \text{ lb} \downarrow = F_{A_y}; F_{A_x} = 0$$

Forces on Gear *C*:

$$W_{t_C} = \frac{T_C}{R_C} = \frac{1890 \text{ lb·in}}{5.0 \text{ in}} = 378 \text{ lb} \rightarrow = F_{C_x}$$

$$W_{r_C} = W_{t_C} \tan 20° = 138 \text{ lb} \uparrow = F_{C_y}$$

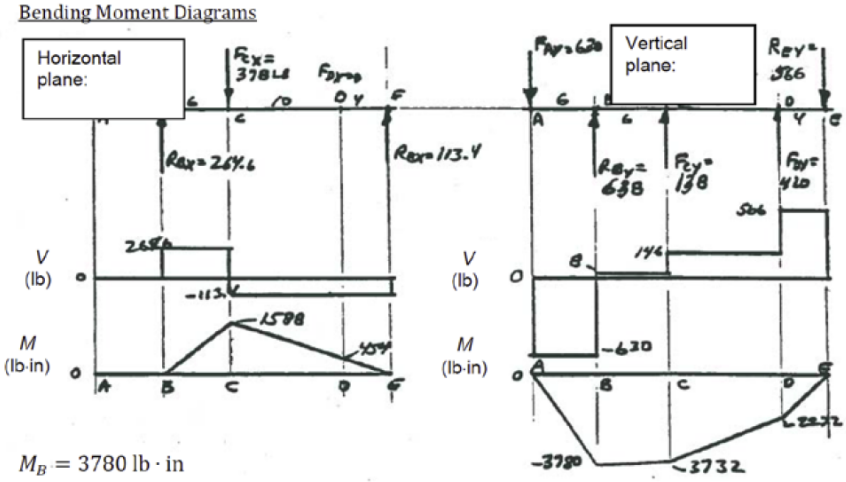


Forces on Sprocket *D*:

$$F_1 = F_D = F_{D_y} = \frac{T_D}{R_D} = \frac{1260}{3} = 420 \text{ lb} \uparrow$$

$$F_{D_x} = 0$$

Bending Moment Diagrams



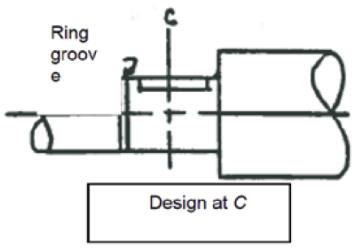
$M_B = 3780$ lb · in

$M_C = \sqrt{1588^2 + 3732^2} = 4056$ lb · in

$M_D = \sqrt{454^2 + 2272^2} = 2317$ lb · in

$T_{C_L} = 3150$ lb · in To left; $K_t = 3.0$ Ring Groove

$T_{C_R} = 1260$ lb · in To right; $K_t = 2.0$ Key



SAE 1170 CD; $s_u = 69$ ksi; $s_y = 57$ ksi

Design at C Specify $D_C = 3.00$ in $C_S = 0.78$ Ok

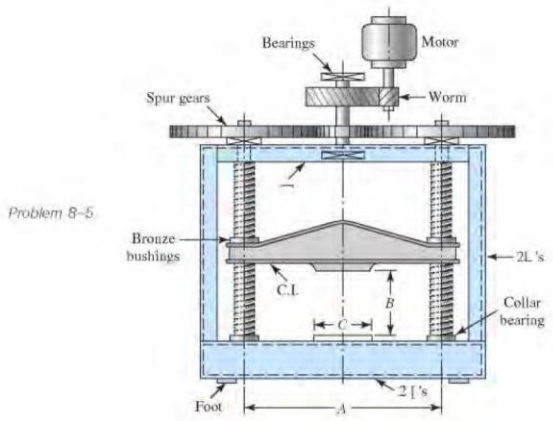
$s_n = 26$ ksi; $S_n' = C_s C_R s_n = (0.85)(0.81)(26) = 17.9$ ksi

$$D_{C_L} = \left(\frac{32(3)}{\pi} \sqrt{\left(\frac{3.0(4056)}{17\,900} \right)^2 + \frac{3}{4}\left(\frac{3150}{51\,000} \right)^2} \right)^{1/3} = 2.75 \text{ in}$$

Increase by 6% at groove $D \approx 1.06(2.75) = 2.92$ in

Note: I believe the 51000 should 57000psi for the yield strength of the SAE 1170 CD.

Problem 2. Adapted from **Shigley, Problem 8-6**.



The press shown above has a rated load of 5000 lbf. The twin screws have Acme threads, a diameter of 2 in, and a pitch of $\frac{1}{4}$ in. Coefficients of friction are 0.05 for the threads and 0.08 for the collar bearings. Collar diameters are 3.5 in. The gears have an efficiency of 95 percent and a speed ratio of 60:1. A slip clutch, on the motor shaft, prevents overloading. The full-load motor speed is 1720 rev/min.

Part 1: To be done by students.

(*a*) When the motor is turned on, how fast will the press head move?
(*b*) What should be the horsepower rating of the motor?

Screws rotate at an angular rate of

$$n = \frac{1720}{60} = 28.67 \text{ rev/min}$$

(a) The lead is 0.25 in, so the linear speed of the press head is

$V = 28.67(0.25) = 7.17$ in/min *Ans.*

(b) $F = 2500$ lbf/screw

$$d_m = 2 - 0.25 / 2 = 1.875 \text{ in}$$
$$\sec \alpha = 1 / \cos(29^\circ / 2) = 1.033$$

Eq. (8-5):

$$T_R = \frac{2500(1.875)}{2}\left(\frac{0.25 + \pi(0.05)(1.875)(1.033)}{\pi(1.875) - 0.05(0.25)(1.033)}\right) = 221.0 \text{ lbf} \cdot \text{in}$$

Eq. (8-6):

$$T_c = 2500(0.08)(3.5 / 2) = 350 \text{ lbf} \cdot \text{in}$$
$$T_{total} = 350 + 221.0 = 571 \text{ lbf} \cdot \text{in/screw}$$
$$T_{motor} = \frac{571(2)}{60(0.95)} = 20.04 \text{ lbf} \cdot \text{in}$$
$$H = \frac{Tn}{63\,025} = \frac{20.04(1720)}{63\,025} = 0.547 \text{ hp} \quad \textit{Ans.}$$

Adapted from **Shigley, Problems 10-3 and 10-5** (8th and 9th Eds.)

A helical compression spring is wound using 0.105-in diameter music wire. The spring has an outside diameter of 1.225 in with plain ground ends, and 12 total coils.

Part 1 – To be demonstrated by the TA

a. What force is needed to compress this spring to closure if the spring is to just reach the yield point upon closure?
b. What is the approximate spring rate for this spring?
c. What should the free length be to ensure that when the spring is compressed solid, the torsional stress does not exceed the yield strength (that is, it is solid safe)?
d. What is the pitch of this spring?
e. Is there a possibility that this spring might buckle in service? (use a safety factor of 5)

Part 2 – To be done by the students

Repeat the analysis above for a helical compression spring with 8-1/2 total coils made of A227 hard-drawn spring steel wire 2 mm diameter and outside diameter of 22 mm. At full compression, the material should remain below yield with a safety factor of 2. The ends are squared and ground and are held in an assembly that allows them to pivot.

Part 1 - To be demonstrated by the TA

a. What force is needed to compress this spring to closure if the spring is to just reach the yield point upon closure?

Some preliminary values:

d	= 0.105 in	spring wire diameter
D_o	= 1.225 in	spring outside diameter
D	= D_o - 2·d/2	mean diameter (distance between coil centres)
	= D_o - d	
	= 1.225 in - 0.105 in	
	= 1.120 in	

The relationship between applied force and stress is

$$\tau = K_s \frac{8FD}{\pi d^3}$$ eq. 10-2 (p. 519)

Where

τ	= shear stress = S_{sy} (shear yield strength) in this case
K_s	= shear stress correction factor (K_W or K_B, see below)
F	= applied force (to be solved for)
D	= mean diameter (above)
d	= wire diameter (given)

Need to find the shear yield strength, S_{sy}

S_{sy} = 0.45S_{ut} See Table 10-6 (p. 526) for music wire

S_{ut} = A/d^m Table 10-4 (p. 525)

A	= 201 kpsi·in^m	Table 10-4 (p. 525)
m	= 0.145	
S_{ut}	= (201 kpsi·in$^{0.145}$)/(0.105 in)$^{0.145}$	
	= 279 kpsi	

S_{sy} = 0.45(279 kpsi)
= 125.4 kpsi

K_s shear stress concentration factor
Can use either $K_s = K_W$ or $K_s = K_B$
As noted in Shigley (bottom of p. 519), K_W and K_B differ by only 1%

so as done in Shigley, use K_B

K_B Bergstrasser factor (eq. 10-5, p. 519)

$$K_B = \frac{4C + 2}{4C - 3}$$

C = D/d eq. 10-1 (p. 519)
= (1.120 in)/(0.105 in)
= 10.67

$$K_B = \frac{4(10.67) + 2}{4(10.67) - 3}$$

$$= 1.126$$

Rearranging 10-2 with $K_s = K_B$ and $\tau = S_{sy}$ in terms of F gives

$$F = \frac{\pi d^3 S_{sy}}{8 K_B D}$$

$$= \frac{\pi (0.105 \text{ in})^3 (125.4 \text{ kpsi})}{8(1.126)(1.120 \text{ in})}$$

$$= 0.0452 \text{ klbf}$$

$$= 45.2 \text{ lbf}$$

A force of 45.2 lbf is required to compress the spring to yield at closure

---

b. What is the approximate spring rate for this spring?

The approximate spring rate, k, is given by eq. 10-9 (p. 520)

$$k \cong \frac{d^4 G}{8 D^3 N} \qquad \text{eq. 10-9 (p. 520)}$$

Where

d & D = diameters as defined above

G = modulus of rigidity
= 11.75 Mpsi — from Table 10-5 (p. 526) for d = 0.105 in music wire

N = N_a = number of active coils

To find N_a, use Table 10-1 (p. 521)

N_a = N_t -1 — for plain and ground ends

N_t = 12 — total number of coils, given

N_a = 12 - 1
= 11

$$k \cong \frac{(0.105 \text{ in})^4 (11.75 \times 10^6 \text{ psi})}{8(1.120 \text{ in})^3 (11)}$$

$$= 11.55 \text{ lbf/in}$$

The approximate spring rate is k = 11.55 lbf/in

c. What should the free length be to ensure that when the spring is compressed solid, the torsional stress does not exceed the yield strength (that is, it is solid safe)?

Knowing the spring rate and the applied force, we can determine how much the spring compresses

$\Delta y = F/k$ F was found in Part a. and k in b.
$= (45.2 \text{ lbf})/(11.55 \text{ lbf/in})$
$= 3.91 \text{ in}$

At maximum compression, the spring is at its solid length, L_s

$L_s = dN_t$ from inspection (or Table 10-1, p. 521)
$= (0.105 \text{ in})(12)$
$= 1.26 \text{ in}$

The free length is just the solid length plus the deflection required to bring the spring to its solid length

$L_0 = L_s + \Delta y$
$= 1.26 \text{ in} + 3.91 \text{ in}$
$= 5.17 \text{ in}$

The free length should be 5.17 in

d. What is the pitch of this spring?

Pitch, p, is distance between coils, paying attention to end conditions. For a plain and ground spring, Table 10-1 (p. 521) gives the pitch

$p = L_o/(N_a + 1)$
$= L_o/N_t$
$= (5.17 \text{ in}) / 12$
$= 0.431 \text{ in}$

The pitch is 0.431 in

e. Is there a possibility that this spring might buckle in service (using a safety factor of 5)?

Buckling is a concern if the free length exceeds a critical value. (The more slender a spring is, the more likely it is to buckle.)

To avoid buckling, $L_0 < L_{o|crit}$. With a safety factor of n, $L_o < L_{o|crit}/n$

$L_{o|crit} = 2.63\, D/\alpha$ eq. 10-13 (p. 523)

Where

$L_{o|crit}$ = the critical length for buckling
α = a constant related to the end condition
= 0.5 (fixed ends, From Table 10-2, p. 522)
D = the mean diameter, found above
= 1.120 in

$L_{o|crit}$ = 2.63 (1.120 in)/0.5
= 5.89 in
$L_{o|cirt}/n$ = 5.89 in/5
= 1.18 in

L_o = 5.17 in From Part c.

$L_o > L_{o|crit}/n$ therefore the design should not be considered safe from buckling

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Part 2 – To be done by the students

Repeat the analysis above for a helical compression spring made of A227 hard-drawn spring steel wire 2 mm in diameter and with an outside diameter of 22 mm. At full compression, the material should remain below yield with a safety factor of 2. The ends are squared and ground and are held in an assembly that allows them to pivot. There are 8-1/2 total coils.

a. What force is needed to compress this spring to closure?

In this case, the safety factor of 2 requires that the shear stress in the spring at maximum compression be half of the shear yield strength ($\tau = S_{sy}/2$)

Some preliminary values:

d = 2 mm spring wire diameter
D_o = 22 mm spring outside diameter
D = $D_o - 2 \cdot d/2$ mean diameter (distance between coil centres)
= D_o - d
= 22 mm - 2 mm
= 20 mm

The relationship between applied force and stress is

$$\tau = K_s \frac{8FD}{\pi d^3}$$ eq. 10-2 (p. 519)

Where

τ = shear stress = S_{sy} (shear yield strength) in this case
K_s = shear stress correction factor (K_W or K_B, see below)
F = applied force (to be solved for)
D = mean diameter (above)
d = wire diameter (given)

Need to find the shear yield strength, S_{sy}

S_{sy} = $0.45S_{ut}$ See Table 10-6 (p. 526) for cold-drawn carbon steel

S_{ut} = A/d^m Table 10-4 (p. 525)

A = 1.783 GPa·mm^m Table 10-4 (p. 525)
m = 0.190
S_{ut} = (1.783 GPa·$mm^{0.190}$)/$(2\text{ mm})^{0.190}$
= 1.563 GPa

S_{sy} = 0.45(1.563 GPa)
= 703 MPa

K_s shear stress concentration factor
Can use either $K_s = K_W$ or $K_s = K_B$
As noted in Shigley (bottom of p. 519), K_W and K_B differ by only 1% so as done in Shigley, use K_B

K_B Bergstrasser factor (eq. 10-5, p. 519)

$$K_B = \frac{4C+2}{4C-3}$$

C = D/d eq. 10-1 (p. 519)
= (20 mm)/(2 mm)
= 10.0

$$K_B = \frac{4(10)+2}{4(10)-3}$$
$$= 1.135$$

Rearranging 10-2 in terms of F with $K_s = K_B$ and $\tau = S_{sy}/2$ (the 2 comes from the stated safety factor of 2 required on stresses) gives

$$F = \frac{\pi d^3 S_{sy}/2}{8K_B D}$$
$$= \frac{\pi(0.002\text{ m})^3(703\text{ MPa})/2}{8(1.135)(0.020\text{ m})}$$
$$= 48.6\text{ N}$$

A force of 48.6 N is required to compress the spring to closure while maintaining maximum stress below yield with a safety factor of 2

b. What is the approximate spring rate for this spring?

The approximate spring rate, k, is given by eq. 10-9 (p. 520)

$$k \cong \frac{d^4 G}{8 D^3 N}$$ eq. 10-9 (p. 520)

Where

d & D = diameters as defined above (d = 2 mm = 0.0787 in)
G = modulus of rigidity
= 79.3 GPa from Table 10-5 (p. 526) for d = 0.0787 A227 steel

N = N_a = number of active coils

To find N_a, use Table 10-1 (p. 521)

N_a = N_t -2 for squared and ground ends
N_t = 8.5 total number of coils, given
N_a = 8.5 - 2
= 6.5

$$k \cong \frac{(0.002\ \text{m})^4 (79.3 \times 10^9\ \text{Pa})}{8(0.020\ \text{m})^3 (6.5)}$$

$$= 3050\ \text{N/m}$$

The approximate spring rate is k = 3.05 kN/m

c. What should the free length be to ensure that when the spring is compressed solid, the torsional stress does not exceed the yield strength (that is, it is solid safe)?

Knowing the spring rate and the applied force, we can determine how much the spring compresses

Δy = F/k F was found in Part a. and k in b.
= (48.6 N)/(3050 N/m)
= 0.0159 m

At maximum compression, the spring is at its solid length, L_s

L_s = dN_t from Table 10-1 (p. 521)
= (0.002 m)(8.5)
= 0.017 m

The free length is just the solid length plus the deflection required to bring the spring to its solid length

L_0 = $L_s + \Delta y$
= 0.017 m + 0.0159 m
= 0.0329 m
= 32.9 mm

The free length should be 32.9 mm

d. What is the pitch of this spring?

Pitch, p, is distance between coils, paying attention to end conditions. For a plain and ground spring, Table 10-1 (p. 521) gives the pitch

$p = (L_o - 2d)/N_a$
$= (32.9 \text{ mm} - 2 \cdot 2\text{mm})/6.5$
$= 4.45 \text{ mm}$

The pitch is 4.45 mm

e. Is there a possibility that this spring might buckle in service (using a safety factor of 5)?

Buckling is a concern if the free length exceeds a critical value. (The more slender a spring is, the more likely it is to buckle.)

To avoid buckling, $L_0 < L_{o|crit}$. With a safety factor of n, $L_o < L_{o|crit}/n$

$L_{o|crit} = 2.63\ D/\alpha$ eq. 10-13 (p. 523)

Where

$L_{o|crit}$ = the critical length for buckling

α = a constant related to the end condition
= 1 (pivoting ends, From Table 10-2, p. 522)

D = the mean diameter, found above
= 20 mm

$L_{o|crit} = 2.63\ (20 \text{ mm})/1.0$
$= 52.6 \text{ mm}$
$L_{o|cirt}/n = 52.6 \text{ mm}/5$
$= 10.52 \text{ mm}$

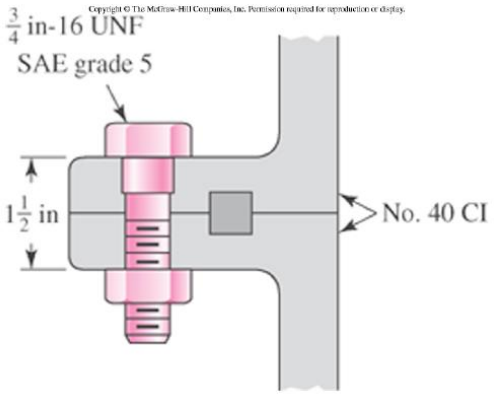
$L_o = 32.9 \text{ mm}$ From Part c.

$L_o > L_{o|crit}/n$ therefore the design should not be considered safe from buckling

Part 1 – To be done at home

a. Show that a 2-½ in bolt has sufficient length.
b. Show that the spring rate of the bolt is 6.78×10^6 lbf/in.
c. Show that the spring rate of the member (including washers) is 14.41×10^6 lbf/in.
d. Show that the stiffness constant of the joint is C = 0.320

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Part 2 – To be done by the students

e. Determine the fatigue factor of safety using the Goodman criterion
f. Determine the fatigue factor of safety using the Gerber criterion
g. Determine the load factor guarding against overproof loading.

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2. Adapted from Shigley, Problem 8-34 (8th Ed.)

The section of the sealed joint shown in the figure is loaded by a repeated force P = 6 kip. The cast iron members have E = 16 Mpsi. Hardened steel washers 0.134 in thick are used under the head and nut. The bolt has been preloaded to a force of F_i = 25 kip each.

Part 1 - To be done at home

a. Show that a 2-½ in bolt has sufficient length.

The bolt must be at least as long as the joint thickness plus the two washers plus the nut.

Joint thickness	= 1.5 in	(given)
Washer height	= 0.134 in	(given)
Nut height	= 41/64 in = 0.641 in	(given)

The required length is

1.5 in + 2(0.134 in) + 0.641 = 2.41 in

A 2.5 in bolt is sufficiently long.

---

b. Show that the spring rate of the bolt is 6.78×10^6 lbf/in.

The spring rate of the bolt is based on the spring rate of the threaded portion and the spring rate of the unthreaded portion. Referring to tutorial 15, the bolt spring rate is

$$k_b = \frac{A_d A_t E}{A_d l_t + A_t l_d}$$

Where

A_t = tensile stress area = 0.373 in^2 (Table 8-2, p. 413, ¾"UNF fastener)

A_d = major diameter area
= $\pi/4 d^2$
= $\pi/4\ (0.75\ in)^2$
= 0.442 in^2

E = 30 Mpsi (Table 8-8 for steel)

l_t = threaded length of grip

l_d = unthreaded length of grip

l_t and l_d must still be determined.

l is the grip length = material thickness plus thickness of two washers

l = 1.5 in + 2(0.134 in)
= 1.768 in

From Table 8-7 (p. 412)

$$L_T = \begin{cases} 2d + \frac{1}{4}\ \text{in} & L \le 6\ \text{in} \\ 2d + \frac{1}{2}\ \text{in} & L > 6\ \text{in} \end{cases}$$

In this case, L = 2.5 in so

L_T = 2d + ¼ in
= 2(0.75 in) + 0.25 in
= 1.75 in

Referring to Table 8-7, the length of the unthreaded portion, l_d, is

l_d = L - L_T (L = 2.5 in given; L_T = 1.75 in from above.)
= 2.5 in - 1.75 in
= 0.75 in

The threaded length in the grip, l, is also shown in Table 8-7,

l_t = l - l_d (l = 1.768 in from above; l_d = 0.75 in from above.)
= 1.768 in - 0.75 in
= 1.018 in

$$k_b = \frac{\left(0.442\ \text{in}^2\right)\left(0.373\ \text{in}^2\right)\left(30\times10^6\ \text{psi}\right)}{\left(0.442\ \text{in}^2\right)\left(1.018\ \text{in}\right)+\left(0.373\ \text{in}^2\right)\left(0.75\ \text{in}\right)}$$

$$= 6.78\times10^6\ \text{lbf/in}$$

The spring rate of the bolts is 6.78 $\times 10^6$ lbf/in.

c. Show that the spring rate of the member (including washers) is 14.41×10^6 lbf/in.

The spring rate is given by the eq. 8-20 (p. 429). Note: the exponential curve fit (eq. 8-23, p. 429) cannot be used in this case because the washer material is different from the other material.

$$k = \frac{0.577\pi Ed}{\ln\dfrac{(1.155t + D - d)(D + d)}{(1.155t + D + d)(D - d)}}$$ eq. 8-20 (p. 429)

For the washers

E = washer elastic modulus (steel)
= 30.0 MPsi (Table 8-8, p. 430)

D = width of the bolt head (Table A-29)
= 1.125 in

Note: the washer diameter is larger than that of the bolt head, but the full washer diameter is not compressed.

d = bolt major diameter (given)
= 0.75 in

t = washer thickness
= 0.134 in

$$k_w = \frac{0.577\pi(30\text{ Mpsi})(0.75\text{ in})}{\ln\dfrac{(1.155(0.134\text{ in}) + 1.125\text{ in} - 0.75\text{ in})(1.125\text{ in} + 0.75\text{ in})}{(1.155(0.134\text{ in}) + 1.125\text{ in} + 0.75\text{ in})(1.125\text{ in} - 0.75\text{ in})}}$$

$$= 153.2\times10^6 \text{ lbf/in}$$

For the flange

E = material elastic modulus (cast iron)
= 14.5 MPsi (Table 8-8, p. 430)

D = this diameter is larger than that used with the washer above due to the spreading of the load (see Fig 8-15, p. 428). The angle of spread is 30° (as noted on p. 428)
= D_{bolt} + 2t·tan(α)
= 1.125 in + 0.134 tan(30°)
= 1.280 in

d = bolt major diameter (given)
= 0.75 in

t = joint thickness (see Figure 8-15, p. 428)
= 0.75 in

$$k_f = \frac{0.577\pi(16.0\text{ Mpsi})(0.75\text{ in})}{\ln\dfrac{(1.155(0.75\text{ in}) + 1.280\text{ in} - 0.75\text{ in})(1.280\text{ in} + 0.75\text{ in})}{(1.155(0.75\text{ in}) + 1.280\text{ in} + 0.75\text{ in})(1.280\text{ in} - 0.75\text{ in})}}$$

$$= 35.5\times10^6 \text{ lbf/in}$$

There are two washers and two sections of material all in series. The net material spring rate is

$$\frac{1}{k_m} = 2\frac{1}{k_w} + 2\frac{1}{k_f}$$
$$= 2\frac{1}{153.3\ \text{Mlbf/in}} + 2\frac{1}{35.5\ \text{Mlbf/in}}$$

The final spring rate is $k_m = 14.40\times10^6$ lbf/in

d. Show that the stiffness constant of the joint is C = 0.320

The joint constant, C, describes the fraction of external load carried by the bolt and is given by eq. (f) on p. 436

$$C = \frac{k_b}{k_b + k_m}$$
$$= \frac{6.78\times10^6\ \text{lbf/in}}{6.78\times10^6\ \text{lbf/in} + 14.40\times10^6\ \text{lbf/in}}$$
$$= 0.320$$

The joint constant is 0.320.

Part 2 – To be done by the students

e. Determine the fatigue factor of safety using the Goodman criterion

For the Goodman criterion without safety factor looks like

$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = 1 \qquad \text{(eq. 6-40, p. 306)}$$

Substituting the relationship for σ_m and noting that the fatigue safety factor, n_f, applies to the alternating component of stress (not the preload)

$$\frac{n_f\sigma_a}{S_e} + \frac{\sigma_i + n_f\sigma_a}{S_{ut}} = 1$$

Rearranging this expression for n gives

$$n_f = \frac{S_e\left(S_{ut} - \sigma_i\right)}{\sigma_a\left(S_{ut} + S_e\right)}$$

As shown in the figure for the problem, the bolt is an SAE grade 5. From Table 8-17 (p. 445), the endurance strength is

S_e = 18.6 kpsi

From Table 8-9 (p. 433) the minimum ultimate tensile strength is

S_{ut} = 120 kpsi

The bolt preload is 25 kip (given) and thus the initial stress in the bolts is

σ_i = F_i/A_t
= (25 kip) / (0.373 in^2)
= 67.0 kpsi

The alternating force magnitude is half of the applied 6 kip force. The bolt experiences a fraction, C, of this load. (C = 0.320, given in Part d.)

σ_a = $CP/(2\ A_t)$
= $(0.320)(6000\ lbf)/(2 \cdot 0.373\ in^2)$
= 2.57 kpsi

$$n_f = \frac{S_e\left(S_{ut} - \sigma_i\right)}{\sigma_a\left(S_{ut} + S_e\right)}$$
$$= \frac{(18.6\ \text{kpsi})(120\ \text{kpsi} - 67.0\ \text{kpsi})}{(2.57\ \text{kpsi})(120\ \text{kpsi} + 18.6\ \text{kpsi})}$$
$$= 2.77$$

The fatigue safety factor for the Goodman criterion is 2.77.

---

f. Determine the fatigue factor of safety using the Gerber criterion

The Gerber criterion safety factor looks can be taken from above

$$\frac{n_f\sigma_a}{S_e} + \left(\frac{\sigma_i + n_f\sigma_a}{S_{ut}}\right)^2 = 1$$

Expanding this expression into a quadratic for n_f gives

$$\sigma_a{}^2 S_e n_f{}^2 + \sigma_a\left(2\sigma_i S_e + S_{ut}{}^2\right)n_f + S_e\left(\sigma_i{}^2 - S_{ut}{}^2\right) = 0$$

Solving this quadratic expression for n_f, taking the positive term, and simplifying gives

$$n_f = \frac{-\left(2\sigma_i S_e + S_{ut}{}^2\right) + \sqrt{\left(2\sigma_i S_e + S_{ut}{}^2\right)^2 - 4S_e\left(\sigma_i{}^2 S_e - S_e S_{ut}{}^2\right)}}{2\sigma_a S_e}$$

The same values of S_e, S_{ut}, σ_i, and σ_a from above can be used (with the units of kpsi omitted)

$$n_f = \frac{-\left(2(67.0)(18.6)+(120)^2\right)+\sqrt{\left(2(18.6)(67.0)+(120)^2\right)^2-4(18.6)^2\left((67.0)^2-(120)^2\right)}}{2(2.57)(18.6)}$$

$$= 4.19$$

The fatigue safety factor using the Gerber Criterion is 4.19

---

g. Determine the load factor guarding against overproof loading.

The proof-strength line on a fatigue diagram is similar to the Goodman line except it intersects both the midrange and alternating stress axes at the proof strength. (see Figure 8-22, p. 449).

The equation describing this line is

$$\frac{n\sigma_a}{S_p}+\frac{\sigma_i+n\sigma_a}{S_p}=1$$

Rearranging this expression for n gives

$$n=\frac{S_p-\sigma_i}{2\sigma_a}$$

From Table 8-9 (p. 433) the proof strength is

S_p = 85 kpsi

The initial and alternating stresses were found above

σ_i = 67.0 kpsi

σ_a = 2.57 kpsi

$$n=\frac{S_p-\sigma_i}{2\sigma_a}$$

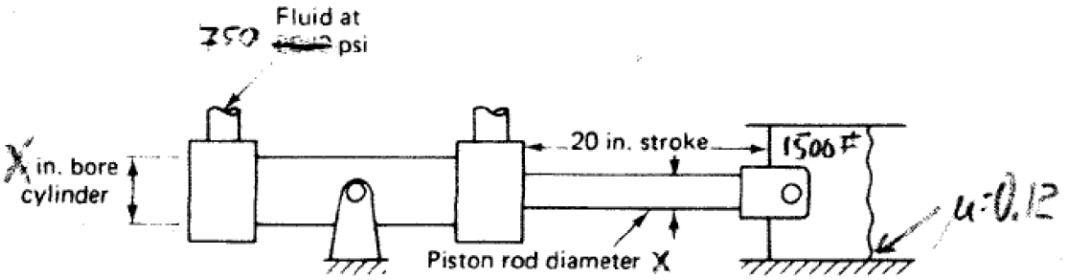
$$=\frac{85\text{ kpsi}-67\text{ kpsi}}{2\cdot 2.57\text{ kpsi}}$$

$$= 3.50$$

The safety factor for overproof loading is 3.50.

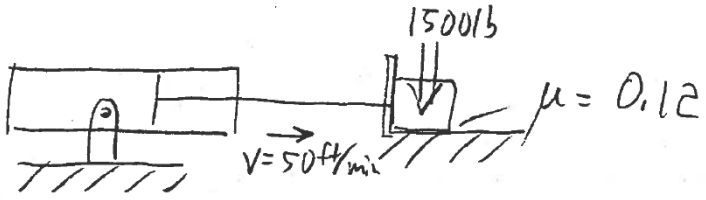
2. **Problem related to Fluid Power Component Sizing**

The following horizontal cylinder is required to move a load of 1500 lbf at a speed of 50 ft/min. over a distance of 20 inches. The friction coefficient has been determined as 0.12 between the load and the floor. In addition, the cylinder must accelerate the load within a distance of 0.75 inches.



a) Including acceleration and friction, determine the required cylinder bore for the system. What is the maximum thrust developed in our cylinder?
b) Select a rod size for the cylinder. Does this fit within the bore selected in part a)? If not, select the minimum required bore for the rod.
c) Size a pump for the system. What is the required flow rate in GPM? What is the required HP for the electric motor?
d) What suggestions can you make in order to reduce the overall cost of the system?

Solution to Problem 2



relief pressure = 750 psi

acceleration distance = 0.75 inch

stroke length = 20 inches

a) total force = W × force factor + friction

V = 50 ft/min S = 0.75

from graph b-2 g = 0.18

(acceleration / deceleration) force = 1500 lbs × 0.18 = 270 lbs

friction force = μW = 0.12(1500 lbs) = 180 lbs

∴ total max force = 270 + 180 = 450 lbs

From Table b-1 @ 750 psi bore = 1" gives 588 lbs max.

b) Determine Rod Size

Use stroke selection table Table b-4 to determine stroke factor

∴ Intermediate Trunnion ∴ load is supported

∴ Stroke Factor = 1.5

Rod length = Actual stroke × Stroke Factor

= 20" × 1.5 = 30 inches

From Piston Rod - Stroke Selection Guide

Thrust = 450 lbs

Basic Length = 30"

value just > 5/8" ∴ use 1" rod diameter

Hence a 1" bore will not work.

Use Table b-5 to select minimum bore with 1" rod size

Bore: 1½ with 1" dia rod.

∴ Cylinder Bore: 1½
Rod Dia: 1"
Stroke: 20"

From Table b-1 max thrust is 1325 lbs at 350 psi

c) Size pump to move 1" rod in 1½ bore at a rate of 50 ft/min

From Table b-5 Cylinder Port Size and Piston Speed

Cylinder End ∅ = 1½ rod ∅ = 0" GPM = 0.92 per 10 ft/min speed

For 50 ft/min speed $Q = 5(0.92) = 4.6$ GPM

double check vol gpm = $\frac{A \times L \times 60}{231 \times \text{time in sec}}$ (page i-9)

$A = \pi r^2 = \pi(1.5/2)^2 = 1.767$ in²

$L = 20$"

$V = 50$ ft/min $= 50 \times 12/60 = 10$"/sec

Total time for stroke = 2 sec.

$\text{vol.(gpm)} = \frac{1.77 \times 20 \times 60}{231 \times 2} = 4.60$ GPM

Pump HP (page i-8)

$Hp = GPM \times psi / 1714 \times \%_{pump}$

Select $\eta = 0.85$ (typical value)

$HP = (4.6\ gpm)(750\ psi)/1714 \times 0.85$

$= 2.36\ hp.$

From table j (C-3) or Table i-5

for $n = 1800$ rpm motor we need to select a motor of 3 HP

d) To reduce costs, it is recommended to reduce working pressure to 500 psi

@ 500 psi max thrust = 885 lb, (Table b-1)

$\therefore$ 885 lb > 450 lbr working load ✓

$HP = (4.6)(500)/1714 \times 0.15$

$= \frac{2}{3}(2.36) = 1.57\ HP.$

@ 400 psi working pressure

max thrust = $\frac{4}{5}(885) = 708$ lbr > 450 ✓

$HP = (4.6)(400)/1714 \times 0.15$

$= 4/5(1.57) = 1.256$

by reducing pressure to 400 psi we can select a $1\frac{1}{2}$ H.P motor rather than 3 HP